

EX.NO: 9

IMPLEMENTATION OF K-MEANS CLUSTERING ALGORITHM

AIM:

To group the IRIS (flower) data points into two different clusters with the help k-means clustering algorithm.

PROCEDURE:

1. Install the necessary packages in R Studio such as “stats”, ”dplyr”, “ggplot2”, “ggfortify”
2. Set the current working directory
3. Save the IRIS dataset in the current working directory
4. Read the dataset in the form of csv file using **read.csv ()**
5. Finally create the model (k=2) with the help of k-means clustering algorithm

SOURCE CODE:

```
# Loading data
data(iris)

# Structre
str(iris)

# Installing Packages
install.packages("Cluster")
install. packages("cluster")

# Loading package
library(ClusterR)
library(cluster)

# Removing initial label of
# Species from original dataset
```

```
iris_1 <- iris[, -5]

# Fitting K-Means clustering Model

# to training dataset

set.seed(240) # Setting seed
kmeans.re <- kmeans(iris_1, centers = 3, nstart = 20) kmeans.re

# Cluster identification for

# each observation kmeans.re$cluster

# Confusion Matrix

cm <- table(iris$Species, kmeans.re$cluster) cm

# Model Evaluation and visualization

plot (iris_1[c("Sepal.Length", "Sepal.Width")])

plot (iris_1[c("Sepal.Length", "Sepal.Width")], col = kmeans.re$cluster)

plot(iris_1[c("Sepal.Length", "Sepal.Width")],col = kmeans.re$cluster,main = "K-means with 3 clusters")

## Plotting cluster centers kmeans.re$centers

kmeans.re$centers[, c("Sepal.Length", "Sepal.Width")]

# cex is font size, pch is symbol

points(kmeans.re$centers[, c("Sepal.Length", "Sepal.Width")], col = 1:3, pch = 8, cex = 3)

## Visualizing clusters y_kmeans <- kmeans.re$cluster

clusplot(iris_1[, c("Sepal.Length", "Sepal.Width")], y_kmeans,

  lines = 0,

  shade = TRUE,

  color = TRUE,

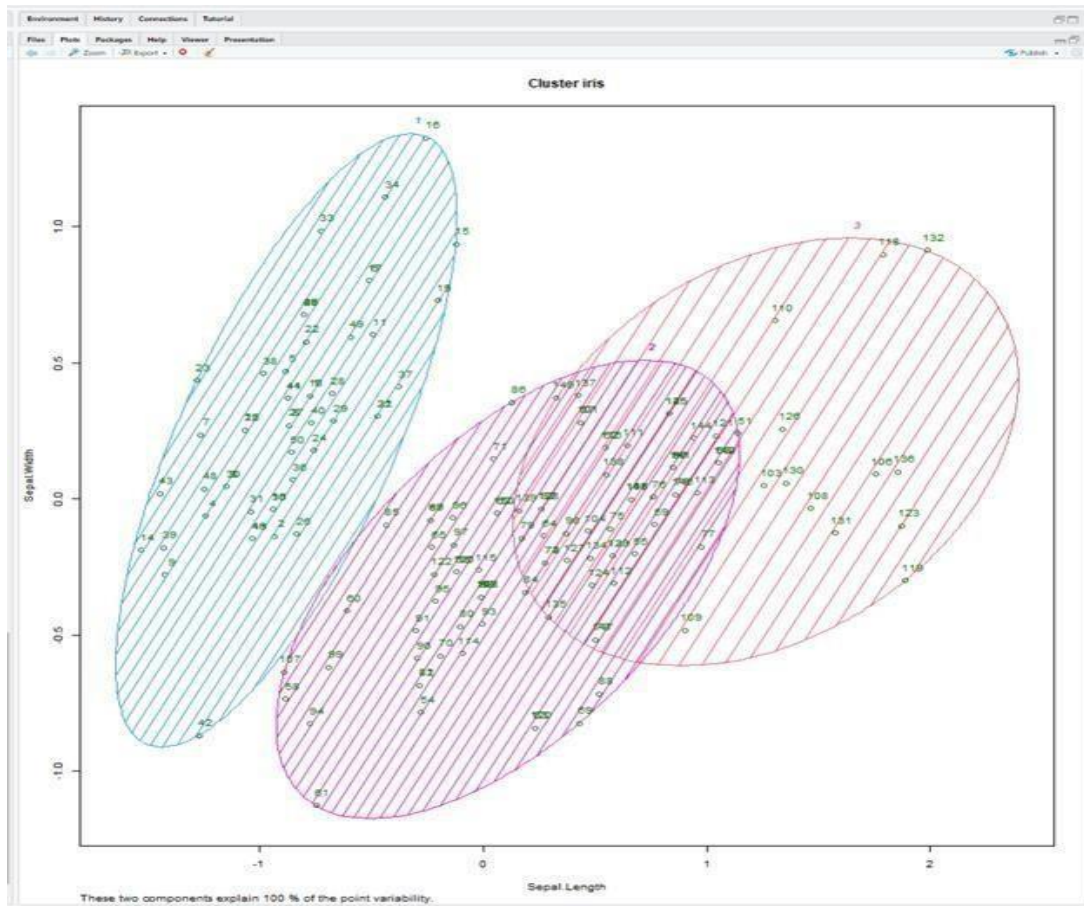
  labels = 2,

  plotchar = FALSE,

  span = TRUE,

  main = paste("Cluster iris"), xlab = 'Sepal.Length', ylab = 'Sepal.Width')
```

OUTPUT:



RESULT:

Thus, the IRIS data points are successfully grouped into two different clusters with the help of kmeans clustering algorithm.